Discrete Random Variables

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Probability distributions**

A discrete random variable X has the probability distribution: $P(X = 1) = 0.2$, $P(X = 2) = 0.3$, $P(X = 3) = k$, $P(X = 4) = 0.1$. Find k, using the fact that probabilities must sum to 1.

$0.2 + 0.3 + k + 0.1 = 1$, so $k = 1 - 0.6 = 0.4$.

2 **[Foundational]** • **Target: Probability distributions**

A discrete random variable Y has the probability distribution: $P(Y = 0) = 0.1$, $P(Y = 1) = k$, $P(Y = 2) = 2k$, $P(Y = 3) = 0.3$. Find k.

$0.1 + k + 2k + 0.3 = 1$, so $3k = 0.6$, giving $k = 0.2$.

3 **[Foundational]** • **Target: Expected value (mean) of a discrete random variable**

A random variable X has distribution $P(X = 1) = 0.2$, $P(X = 2) = 0.5$, $P(X = 3) = 0.3$. Find $E(X)$.

$E(X) = \sum xP(X = x) = 1(0.2) + 2(0.5) + 3(0.3) = 0.2 + 1.0 + 0.9 = 2.1$.

4 **[Foundational]** • **Target: Variance of a discrete random variable**

Using the distribution from the previous question ($E(X) = 2.1$), find $\text{Var}(X)$, using $\text{Var}(X) = E(X^2) - [E(X)]^2$.

$E(X^2) = 1^2(0.2) + 2^2(0.5) + 3^2(0.3) = 0.2 + 2.0 + 2.7 = 4.9$. $\text{Var}(X) = 4.9 - (2.1)^2 = 4.9 - 4.41 = 0.49$.

5 **[Foundational]** • **Target: Variance of a discrete random variable**

Using the distribution $P(Y = 0) = 0.1$, $P(Y = 1) = 0.4$, $P(Y = 2) = 0.5$ ($E(Y) = 1.4$), find $\text{Var}(Y)$, using $\text{Var}(Y) = E(Y^2) - [E(Y)]^2$.

$E(Y^2) = 0^2(0.1) + 1^2(0.4) + 2^2(0.5) = 0 + 0.4 + 2.0 = 2.4$. $\text{Var}(Y) = 2.4 - (1.4)^2 = 2.4 - 1.96 = 0.44$.

6 **[Foundational]** • **Target: Linear transformations of a random variable**

A random variable X has $E(X) = 5$ and $\text{Var}(X) = 3$. Find $E(2X + 4)$ and $\text{Var}(2X + 4)$.

$E(2X + 4) = 2E(X) + 4 = 2(5) + 4 = 14$. $\text{Var}(2X + 4) = 2^2\text{Var}(X) = 4(3) = 12$.

7 **[Foundational]** • **Target: The discrete uniform distribution**

A fair six-sided die is rolled, and X is the number shown. Find $E(X)$ and $\text{Var}(X)$, given that for a discrete uniform distribution on $1, \ldots, n$: $E(X) = \frac{n+1}{2}$ and $\text{Var}(X) = \frac{n^2-1}{12}$.

$n = 6$: $E(X) = \frac{7}{2} = 3.5$. $\text{Var}(X) = \frac{35}{12} \approx 2.92$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: The discrete uniform distribution**

A fair eight-sided die (labeled 1 to 8) is rolled, and X is the number shown. Find $E(X)$ and $\text{Var}(X)$, using the same formulas.

$n = 8$: $E(X) = \frac{9}{2} = 4.5$. $\text{Var}(X) = \frac{63}{12} = 5.25$.

9 **[Intermediate]** • **Target: The discrete uniform distribution**

A fair ten-sided die (labeled 1 to 10) is rolled, and X is the number shown. Find $E(X)$ and $\text{Var}(X)$, using the same formulas.

$n = 10$: $E(X) = \frac{11}{2} = 5.5$. $\text{Var}(X) = \frac{99}{12} = 8.25$.

10 **[Intermediate]** • **Target: Constructing a probability distribution from a real scenario**

A game involves rolling two fair dice and finding the sum X. Find $P(X = 4)$ and $P(X = 7)$.

There are 36 equally likely outcomes. $X = 4$: $(1, 3), (2, 2), (3, 1)$, so $P(X = 4) = \frac{3}{36} = \frac{1}{12}$. $X = 7$: $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$, so $P(X = 7) = \frac{6}{36} = \frac{1}{6}$.

11 **[Intermediate]** • **Target: Constructing a probability distribution from a real scenario**

Using the same game of rolling two fair dice and finding the sum X, find $P(X = 9)$ and $P(X = 12)$.

$X = 9$: $(3, 6), (4, 5), (5, 4), (6, 3)$, so $P(X = 9) = \frac{4}{36} = \frac{1}{9}$. $X = 12$: $(6, 6)$, so $P(X = 12) = \frac{1}{36}$.

12 **[Intermediate]** • **Target: A well-designed word problem: a raffle ticket game**

This question has two parts. A raffle sells 200 tickets for 5 dollars each. One ticket wins a prize of 300 dollars, and two tickets each win a prize of 50 dollars. Let X be the net profit (prize won minus the 5-dollar ticket cost) for a single ticket. (a) Write the probability distribution of X. (b) Find $E(X)$, and interpret whether buying a ticket is a fair game.

(a) $X = 295$ (win big prize) with $P = \frac{1}{200}$; $X = 45$ (win small prize) with $P = \frac{2}{200}$; $X = -5$ (no prize) with $P = \frac{197}{200}$. (b) $E(X) = 295\left(\frac{1}{200}\right) + 45\left(\frac{2}{200}\right) + (-5)\left(\frac{197}{200}\right) = \frac{295 + 90 - 985}{200} = \frac{-600}{200} = -3$. Since $E(X) = -3 \neq 0$, the game is not fair — on average, a player loses 3 dollars per ticket.

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13 **[Intermediate]** • **Target: The discrete uniform distribution**

A discrete random variable X follows a binomial distribution $X \sim B(15, 0.35)$. (a) Find $P(X = 5)$. (b) Find $P(X \geq 4)$. (c) Calculate the mean $E(X)$ and standard deviation σ of X.

(a) $P(X = 5) = \binom{15}{5}(0.35)^5(0.65)^{10} = 3003 \times (0.005252) \times (0.013463) \approx 0.2123$. (b) $P(X \geq 4) = 1 - P(X \leq 3)$. Using GDC `binomcdf(15, 0.35, 3)`: $P(X \leq 3) \approx 0.1727$. $P(X \geq 4) = 1 - 0.1727 = 0.8273 \quad (82.7\%)$. (c) Mean: $E(X) = np = 15 \times 0.35 = 5.25$. Variance: $\text{Var}(X) = np(1-p) = 15 \times 0.35 \times 0.65 = 3.4125$. Standard deviation: $\sigma = \sqrt{3.4125} \approx 1.847 \approx 1.85$.

14 **[Intermediate]** • **Target: Constructing a probability distribution from a real scenario**

A random variable X has distribution $\text{Po}(\lambda)$. A single observation of X is taken to test $H_0 : \lambda = 6$ against $H_1 : \lambda < 6$ at the 5% level. (a) Find the critical region. (b) State the actual significance level. (c) Find the probability of a Type II error if the true value of λ is 3.

(a) Under H_0, $X \sim \text{Po}(6)$. Calculate cumulative probabilities: $P(X = 0) = e^{-6} \approx 0.00248$ $P(X \leq 1) = e^{-6}(1 + 6) = 7e^{-6} \approx 0.01735$ $P(X \leq 2) = e^{-6}(1 + 6 + 18) = 25e^{-6} \approx 0.06197$ Since $P(X \leq 1)0.05$, the critical region is $X \leq 1$ (i.e. $X = 0$ or 1). (b) Actual significance level = $P(X \leq 1 \mid \lambda = 6) = 7e^{-6} \approx 0.0174$ (or 1.74%). (c) Type II error occurs when we fail to reject H_0 when $\lambda = 3$. We fail to reject when.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Cumulative distribution function** • **Cross-Lesson: 13.01 Newton's Laws**

A continuous random variable X has PDF given by $f(x) = \frac{k}{x^2}$ for $1 \leq x \leq 4$, and 0 elsewhere. (a) Show that $k = \frac{4}{3}$. (b) Find the median of X.

(a) Integrate over the support and equate to 1: $\int_1^4 \frac{k}{\sqrt{2}}\,dx = k[-\frac{1}{x}]_1^4 = k(-\frac{1}{4} - (-1)) = \frac{3}{4}k = 1 \Rightarrow k = \frac{4}{3}$ (b) Find the CDF for $1 \leq x \leq 4$: $F(x) = \int_1^x \frac{4}{3t^2}\,dt = \frac{4}{3}(1 - \frac{1}{x})$ Set $F(m) = 0.5$ to find the median m.

16 **[Challenging]** • **Target: A well-designed word problem: a raffle ticket game** • **Cross-Lesson: 14.01 Energy, Work & Power**

The number of accidents on a highway follows a Poisson distribution with mean 1.6 per week. (a) Find the probability of at least 3 accidents in a 2-week period. (b) Given that at least one accident occurs in a week, find the probability that there are exactly 2 accidents in that week.

(a) In a 2-week period, the parameter scales to $\lambda = 1.6 \times 2 = 3.2$:

$P(X \geq 3) = 1 - P(X \leq 2) = 1 - e^{-3.2}(1 + 3.2 + \frac{3.2^2}{2}) = 1 - 9.32e^{-3.2} \approx 1 - 0.3799 = 0.620$ (3 s. f.) (b) For 1 week, $\lambda = 1.6$. Use conditional probability:

$P(X = 2 \mid X \geq 1) = \frac{P(X=2)}{P(X \geq 1)} = \frac{\frac{e^{-1.6}(1.6)^2}{2}}{1 - e^{-1.6}} = \frac{1.28e^{-1.6}}{1 - e^{-1.6}} \approx \frac{0.2584}{0.7981} \approx 0.324$ (3 s. f.)

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17 [Challenging] • Target: Probability distributions • Cross-Lesson: 15.01 Representation of Data

The lifespan X in hours of an electronic component is normally distributed with mean $\mu = 1200$ hours and standard deviation $\sigma = 150$ hours. (a) Find the probability that a randomly selected component lasts more than 1350 hours. (b) In a batch of 500 components, estimate how many will last between 1000 and 1400 hours. (c) The manufacturer guarantees the top 10% longest-lasting components. Determine the minimum lifespan required to receive this guarantee.

(a) Standardize: $Z = \frac{1350 - 1200}{150} = \frac{150}{150} = 1.0$. $P(X > 1350) = P(Z > 1.0) = 1 - \Phi(1.0) \approx 1 - 0.8413 = 0.1587$ (15.9%). (b) For $1000 \le X \le 1400$: $Z_1 = \frac{1000 - 1200}{150} = -\frac{200}{150} = -1.3333$. $Z_2 = \frac{1400 - 1200}{150} = \frac{200}{150} = 1.3333$. Using GDC `normalcdf(-1.3333, 1.3333, 0, 1)`: $P \approx 0.8176$. Expected number: $500 \times 0.8176 \approx 408.8 \approx 409$ components. (c) We require.

18 [Challenging] • Target: Expected value (mean) of a discrete random variable • Cross-Lesson: 16.01 Permutations & Combinations

The masses, x grams, of 50 packets of tea gave $\sum x = 5090$ and $\sum x^2 = 518380$. (a) Calculate unbiased estimates of the population mean and variance. (b) Calculate a 95% confidence interval for the population mean.

(a) Sample mean: $\bar{x} = \frac{5090}{50} = 101.8$ g Unbiased sample variance: $s^2 = \frac{1}{50-1}(518380 - 50(101.8)^2) = \frac{1}{49}(518380 - 518162) = \frac{218}{49} \approx 4.449$ g^2 (b) Standard error: $\text{SE} = \sqrt{\frac{s^2}{n}} = \sqrt{\frac{4.449}{50}} \approx 0.2983$. $\bar{x} \pm 1.96(0.2983) = 101.8 \pm 0.585 = (101.2,\ 102.4)$ g (1 d. p.)

19 [Challenging] • Target: Variance of a discrete random variable • Cross-Lesson: 18.01 Binomial Distribution

Practice 2.2: The graph of $y = f(x)$ undergoes three consecutive transformations: (1) a horizontal stretch by scale factor 3, (2) a reflection in the y-axis, and (3) a translation by vector $(4;\ -5)$. Write down the equation of the resulting function $g(x)$ in terms of f.

1. Horizontal stretch by factor 3: $x \to \frac{x}{3}$, giving $y = f\left(\frac{x}{3}\right)$. 2. Reflection in y-axis: $x \to -x$, giving $y = f\left(-\frac{x}{3}\right)$. 3. Translation by $(4;\ -5)$: replace x by $x - 4$ and subtract 5: $g(x) = f\left(-\frac{x-4}{3}\right) - 5 = f\left(\frac{4-x}{3}\right) - 5$.

20 [Challenging] • Target: Linear transformations of a random variable • Cross-Lesson: 19.01 Normal Distribution

$f(x) = cx^2$ for $0 \le x \le 3$, zero otherwise. Find c, $E(X)$, and $P(1 \le X \le 2)$.

$\int_0^3 cx^2\,dx = 9c = 1 \Rightarrow c = \frac{1}{9}$. $E(X) = \frac{1}{9}\int_0^3 x^3\,dx = \frac{1}{9} \cdot \frac{81}{4} = \frac{9}{4}$. $P(1 \le X \le 2) = \frac{1}{9}\left[\frac{x^3}{3}\right]_1^2 = \frac{1}{27}(8 - 1) = \frac{7}{27}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Probability distributions

A discrete random variable X is uniformly distributed on $\{1, 2, \ldots, 7\}$. Find $E(X)$.

a. 9/2

b. 3

c. **4** ✓

d. 5

$E(X) = \frac{n+1}{2} = \frac{7+1}{2} = 4$

22 [Foundational] • Target: Expected value (mean) of a discrete random variable

Two independent random samples, each of size n, are taken from a normal population with mean μ and known standard deviation σ. From each sample, a symmetric 90% confidence interval for μ is constructed. Given that $\Phi(1.645) = 0.9500$ and $\Phi(2.326) = 0.9900$, where Φ denotes the standard normal cumulative distribution function, find the probability that these two confidence intervals overlap.

a. **0.980** ✓

b. 0.900

c. 0.810

d. 0.990

1. **Margin of error for each confidence interval:** For a symmetric 90% confidence interval with known σ, the critical value is $z = 1.645$. Each interval has half-width: $E = 1.645\frac{\sigma}{\sqrt{n}}$ The two intervals are $I_1 = [\bar{X}_1 - E, \bar{X}_1 + E]$ and $I_2 = [\bar{X}_2 - E, \bar{X}_2 + E]$. 2. **Condition for interval overlap:** Two intervals of identical half-width E overlap if and only if the distance between their centres does not exceed the sum of their half-widths.

23 [Foundational] • Target: Variance of a discrete random variable

A wire of length 4 m is bent to form a rectangle with adjacent sides of lengths X m and $(2 - X)$ m, where X is a continuous random variable with probability density function $f(x) = \begin{cases} \frac{3}{8}x^2, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$ Given that the area of the rectangle is at least $\frac{3}{4}$ m^2, find the probability that the length of the shorter side is at most $\frac{2}{3}$ m.

a. $\frac{127}{351}$ ✓

b. $\frac{74}{351}$

c. $\frac{37}{351}$

d. $\frac{127}{640}$

Let A denote the area of the rectangle and $L_{\min}$ denote the length of the shorter side. 1. **Determine the condition for $A \ge \frac{3}{4}$:** The area is given by $A = X(2 - X) = 2X - X^2$.
$2X - X^2 \ge \frac{3}{4} \Leftrightarrow X^2 - 2X + \frac{3}{4} \le 0 \Leftrightarrow (X - \frac{1}{2})(X - \frac{3}{2}) \le 0$ Thus, $A \ge \frac{3}{4}$ corresponds to the interval $E_1 = \left[\frac{1}{2}, \frac{3}{2}\right]$. Using the cumulative distribution function.

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24 [Foundational] • Target: Linear transformations of a random variable

The continuous random variable X has probability density function given by $f(x) = \{k(4 - x^2), 0 \le x \le 2, ; 0, \text{otherwise}, \text{ where } k \text{ is a constant. Find the value of } k.$

a. $\frac{3}{16}$ ✓

b. $\frac{1}{4}$

c. $\frac{3}{8}$

d. $\frac{16}{3}$

For $f(x)$ to be a valid probability density function, the total area under the curve must equal 1: $\int_{-\infty}^{\infty} f(x)\,dx = 1$ Substitute the non-zero definition of $f(x)$: $\int_0^2 k(4 - x^2)\,dx = 1$ Factor out the constant k and integrate term by term: $k\left[4x - \frac{x^3}{3}\right]_0^2 = 1$ Evaluate at the upper and lower limits: $k\left(\left(4(2) - \frac{2^3}{3}\right) - 0\right) = 1$ $k\left(8 - \frac{8}{3}\right) = 1$.

25 [Foundational] • Target: The discrete uniform distribution

A continuous random variable X has probability density function given by $f(x) = \frac{3}{8}x^2$ for $0 \le x \le 2$, and $f(x) = 0$ otherwise. Find $\mathrm{P}(1 \le X \le 2)$.

a. $\frac{7}{8}$ ✓

b. $\frac{1}{8}$

c. $\frac{3}{4}$

d. $\frac{1}{2}$

To find the probability $\mathrm{P}(1 \le X \le 2)$, we integrate the probability density function $f(x)$ over the interval $[1, 2]$: $\mathrm{P}(1 \le X \le 2) = \int_1^2 \frac{3}{8}x^2\,dx$ Find the antiderivative: $\int \frac{3}{8}x^2\,dx = \frac{3}{8} \cdot \frac{x^3}{3} = \frac{x^3}{8}$ Evaluate between the limits $x = 1$ and $x = 2$: $\left[\frac{x^3}{8}\right]_1^2 = \frac{2^3}{8} - \frac{1^3}{8} = \frac{8}{8} - \frac{1}{8} = \frac{7}{8}$ Common errors include calculating t.

26 [Foundational] • Target: Constructing a probability distribution from a real scenario

The continuous random variable X has probability density function $f(x)$ defined by \[f(x) = \begin{cases} kx & 0 \le x 1 \mid X < 2.5) \).

a. $\frac{9}{11}$ ✓

b. $\frac{10}{11}$

c. $\frac{3}{4}$

d. $\frac{9}{10}$

Step 1: Determine the constant k using the total probability property $\int_{-\infty}^{\infty} f(x)\,dx = 1$. $\int_0^2 kx\,dx + \int_2^3 k(6 - 2x)\,dx = 1$ Evaluating each integral: $\int_0^2 kx\,dx = k\left[\frac{x^2}{2}\right]_0^2 = 2k$ $\int_2^3 k(6 - 2x)\,dx = k[6x - x^2]_2^3 = k((18 - 9) - (12 - 4)) = k(9 - 8) = k$ Thus, $2k + k = 3k = 1 \Rightarrow k = \frac{1}{3}$. Step 2: Find $\mathrm{P}(X1 \mid X1)$ instead of.

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27 [Foundational] • Target: Cumulative distribution function

The random variable X has a Poisson distribution with mean $\lambda > 0$. Given that $3\,\mathrm{P}(X = 2) = \mathrm{P}(X = 4)$, find the value of λ.

a. 6 ✓

b. 36

c. 12

d. $\sqrt{6}$

For a Poisson distribution $X \sim \mathrm{Po}(\lambda)$, the probability mass function is given by: $\mathrm{P}(X = r) = \frac{e^{-\lambda}\lambda^r}{r!}$ Substitute $r = 2$ and $r = 4$ into the given equation $3\,\mathrm{P}(X = 2) = \mathrm{P}(X = 4)$: $3\left(\frac{e^{-\lambda}\lambda^2}{2!}\right) = \frac{e^{-\lambda}\lambda^4}{4!}$ Since $e^{-\lambda} > 0$ and $\lambda > 0$, divide both sides by $e^{-\lambda}\lambda^2$: $\frac{3}{2} = \frac{\lambda^2}{24}$ Multiply both sides by 24: $\lambda^2 = 36$ Since.

28 [Foundational] • Target: A well-designed word problem: a raffle ticket game

The length, X mm, of components manufactured by an industrial process is normally distributed with mean μ mm and standard deviation σ mm. A quality assurance engineer tests the null hypothesis $H_0 : \mu = 20.0$ against the alternative hypothesis $H_1 : \mu > 20.0$ at the 5% significance level using an initial random sample of size $n = 25$. The critical region for the sample mean $\bar{X}$ is found to be $\bar{X} \geq 20.658$. Using the standard normal critical value $z_{0.05} = 1.645$: 1. Find the value of σ. 2. Hence, determine the minimum sample size, m, required so that in a subsequent test of $H_0 : \mu = 20.0$ against $H_1 : \mu > 20.0$ at the same 5% significance level, the probability of committing a Type II error when the true mean is $\mu = 21.0$ is at most 0.05.

a. **44** ✓

b. 43

c. 11

d. 22

We solve the problem in two stages: **Step 1: Determine σ** Under H_0, the distribution of the sample mean is $\bar{X} \sim N\left(20.0, \frac{\sigma^2}{25}\right)$. The critical value for a upper one-tailed test at the 5% level corresponds to $z = 1.645$: $\frac{20.658 - 20.0}{\frac{\sigma}{\sqrt{25}}} = 1.645$ $\frac{0.658}{\frac{\sigma}{5}} = 1.645 \Rightarrow \frac{3.29}{\sigma} = 1.645 \Rightarrow \sigma = \frac{3.29}{1.645} = 2.0$ mm **Step 2: Determine the minimum sample size m** For.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Probability distributions

Flaws in an optical fiber occur according to a Poisson process with a constant mean rate of λ flaws per kilometer. An automated quality-assurance test inspects two independent sections of fiber. The first section has length 1 km and the second section has length 2 km. Let X and Y denote the number of flaws observed in the first and second sections, respectively. Given that the total number of flaws observed across both sections is $X + Y = 3$, find the conditional probability $\mathrm{P}(X \geq 1 \mid X + Y = 3)$.

a. $\frac{19}{27}$ ✓ **b.** $\frac{8}{27}$

c. $\frac{7}{8}$ **d.** $\frac{2}{3}$

Let $X \sim \mathrm{Po}(\lambda)$ and $Y \sim \mathrm{Po}(2\lambda)$ be independent Poisson random variables. The sum $T = X + Y$ follows a Poisson distribution with parameter $\lambda + 2\lambda = 3\lambda$. The conditional distribution of X given $X + Y = n$ is binomial $\mathrm{B}\left(n, \frac{\lambda}{\lambda + 2\lambda}\right) = \mathrm{B}\left(n, \frac{1}{3}\right)$. Here, $n = 3$, so $X \mid (X + Y = 3) \sim \mathrm{B}\left(3, \frac{1}{3}\right)$. We want to find.

30 [Intermediate] • Target: Expected value (mean) of a discrete random variable

The random variable X has a Poisson distribution with parameter λ, where $\lambda > 0$. Given that $\mathrm{P}(X = 3) = \mathrm{P}(X = 4)$, find the value of λ.

a. **4** ✓ **b.** 3

c. 12 **d.** $\frac{4}{3}$

For a random variable $X \sim \mathrm{Po}(\lambda)$, the probability mass function is given by: $\mathrm{P}(X = x) = \frac{\mathrm{e}^{-\lambda}\lambda^{x}}{x!}, \quad x = 0, 1, 2, \ldots$ Given $\mathrm{P}(X = 3) = \mathrm{P}(X = 4)$, we substitute $x = 3$ and $x = 4$: $\frac{\mathrm{e}^{-\lambda}\lambda^{3}}{3!} = \frac{\mathrm{e}^{-\lambda}\lambda^{4}}{4!}$ Since $\mathrm{e}^{-\lambda} > 0$ and $\lambda > 0$, we can divide both sides by $\mathrm{e}^{-\lambda}\lambda^{3}$: $\frac{1}{3!} = \frac{\lambda}{4!}$ Multiplyi.

31 [Intermediate] • Target: Variance of a discrete random variable

A random sample of 10 values of a variable x is taken. The results are summarised as follows: $\sum x = 50, \quad \sum x^2 = 322$ Calculate an unbiased estimate of the population variance.

a. **8** ✓ **b.** 7.2

c. 2.83 **d.** 32.2

To find an unbiased estimate of the population variance s^2, we use the formula: $s^2 = \frac{1}{n-1}\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)$ Given $n = 10$, $\sum x = 50$, and $\sum x^2 = 322$: $\frac{(\sum x)^2}{n} = \frac{50^2}{10} = \frac{2500}{10} = 250$ Substitute these values into the formula: $s^2 = \frac{1}{10-1}(322 - 250) = \frac{1}{9} \times 72 = 8$ - The distractor 7.2 results from dividing by $n = 10$ instead of $n - 1 = 9$, which gives the biased sample variance rather than the unbiase.

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32 [Intermediate] • Target: Linear transformations of a random variable

A continuous random variable X has PDF $f(x) = \frac{2}{2^2}x$ for $0 \le x \le 2$. Find the median of X in exact form.

a. $\frac{2\sqrt{2}}{2}$ ✓

b. $\frac{2\sqrt{2}}{4}$

c. $\frac{2}{2}$

d. $2\sqrt{2}$

$F(m) = (\frac{m}{2})^2 = \frac{1}{2} \Rightarrow m = \frac{2}{\sqrt{2}} = \frac{2\sqrt{2}}{2}$.

33 [Intermediate] • Target: The discrete uniform distribution

A continuous random variable X has PDF $f(x) = kx^1$ for $0 \le x \le 4$ (and 0 otherwise). Find k.

a. $\frac{5}{24}$

b. $\frac{1}{4}$

c. $\frac{1}{16}$

d. $\frac{1}{8}$ ✓

$\int_0^4 kx^1 dx = k\frac{4^2}{2} = 1 \Rightarrow k = \frac{2}{16} = \frac{1}{8}$.

34 [Intermediate] • Target: Constructing a probability distribution from a real scenario

A continuous random variable X has PDF $f(x) = \frac{2}{4^2}x^1$ for $0 \le x \le 4$. Find $\text{Var}(X)$.

a. $\frac{17}{9}$

b. $\frac{8}{9}$ ✓

c. $\frac{8}{3}$

d. 8

$E(X) = \frac{8}{3}$, $E(X^2) = 8$. $\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{8}{9}$.

35 [Intermediate] • Target: Cumulative distribution function

A binomial random variable has $n = 200$, $p = \frac{1}{50}$, and is approximated by a Poisson distribution. Find the parameter λ of the approximating Poisson distribution.

a. 50

b. **4** ✓

c. 200

d. 5

$\lambda = np = 200 \times \frac{1}{50} = 4$.

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36 [Intermediate] • Target: A well-designed word problem: a raffle ticket game

The graph of the quadratic function $f(x) = x^2 - 8x + k$ is tangent to the x-axis. The graph of the linear function $g(x) = -x + 1$ is also shown. Find the value of k.

a. **16** ✓

b. 8

c. 4

d. 2

For the quadratic function $f(x) = x^2 - 8x + k$ to be tangent to the x-axis, its discriminant must be zero. The discriminant of a quadratic $ax^2 + bx + c$ is $b^2 - 4ac$. In this case, $a = 1$, $b = -8$, and $c = k$. So, $(-8)^2 - 4(1)(k) = 0$, which means $64 - 4k = 0$. Solving for k, we get $4k = 64$, so $k = 16$. The value of k is 16. The graph of $g(x)$ is not needed to solve for k; it is provided to ensure the problem adheres to the problem archetype of using graphical information and potentially having dist.

37 [Intermediate] • Target: Probability distributions

Find the distance between the points $(7, 4)$ and $(16, -8)$.

a. 21

b. 14

c. **15** ✓

d. 16

$d = \sqrt{(9)^2 + (12)^2} = \sqrt{225} = 15$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Expected value (mean) of a discrete random variable • Cross-Lesson: 16.01 Permutations & Combinations

The rational function f is defined by $f(x) = \frac{2x^4 - 3x^3 + kx^2 - 8x + 3}{(x - 2)(x^2 + 1)}$, where k is a real constant. Given that the series expansion of $f(x)$ in ascending powers of x, valid for sufficiently small $|x|$, has no term in x^2, find the value of k.

a. $\frac{25}{4}$ ✓

b. $\frac{17}{4}$

c. $-\frac{25}{4}$

d. $\frac{115}{12}$

To find the series expansion of $f(x)$, we first decompose $f(x)$ into partial fractions. **Step 1: Algebraic Division** The degree of the numerator is 4 and the degree of the denominator is $1 + 2 = 3$. Because the numerator has a higher degree than the denominator, $f(x)$ is an improper algebraic fraction and polynomial division must be carried out first. Expanding the denominator:
$(x - 2)(x^2 + 1) = x^3 - 2x^2 + x - 2$. Dividing $2x^4 - 3x^3 + kx^2 - 8x + 3$ by $x^3 - 2x^2 + x - 2$.

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39 [Challenging] • Target: Variance of a discrete random variable • Cross-Lesson: 18.01 Binomial Distribution

The function f is defined by $f(x) = x^2 - 6x + 13 \quad \text{for } x \geq 3$. Determine the complete set of real values of the constant k for which the equation $f^{-1}(x) = |2x - k|$ has exactly one real solution.

a. $k = \frac{39}{8}$ **or** $5 < k < 11$ ✓

b. $k = \frac{39}{8}$ or $5 \leq k < 11$

c. $\frac{39}{8} < k \leq 5$ or $k \geq 11$

d. $k = \frac{39}{8}$ only

1. **Find $f^{-1}(x)$ and its domain:** Completing the square gives $f(x) = (x-3)^2 + 4$. For the domain $x \geq 3$, $f(x)$ is strictly increasing with range $[4, \infty)$. Setting $y = (x-3)^2 + 4$ with $x \geq 3$ yields $x - 3 = \sqrt{y-4}$, so $f^{-1}(x) = 3 + \sqrt{x-4}, \quad \text{with domain } x \geq 4$. 2. **Set up the equation $f^{-1}(x) = |2x - k|$:** Let $u = \sqrt{x-4} \geq 0$, which implies $x = u^2 + 4 \geq 4$. The equation becomes $3 + u = |2(u^2+4) - k| = |2u^2 + 8 - k|$. 3. **Analyz.

40 [Challenging] • Target: Linear transformations of a random variable • Cross-Lesson: 19.01 Normal Distribution

A quality control protocol uses a two-stage hypothesis test to monitor the rate of flaws, λ per metre, in the production of high-grade optical fibre. The number of flaws in a length of fibre follows a Poisson distribution. The manufacturer tests: $H_0 : \lambda = 1 \quad \text{against} \quad H_1 : \lambda > 1$. The two-stage procedure operates as follows: - **Stage 1:** A $1-\text{metre}$ sample is inspected, and the number of flaws X_1 is recorded. If $X_1 \geq 3$, H_0 is rejected immediately. If $X_1 = 0$, H_0 is accepted immediately. If $X_1 \in \{1, 2\}$, the test proceeds to Stage 2. - **Stage 2:** A second, independent $1-\text{metre}$ sample is inspected, and the number of flaws X_2 is recorded. If $X_1 + X_2 \geq 4$, H_0 is rejected; otherwise, H_0 is accepted. Given that the true flaw rate is $\lambda = 2$ per metre, find the probability that a Type II error is committed.

a. $e^{-2} + 16e^{-4} \approx 0.428$ ✓

b. $16e^{-4} \approx 0.293$

c. $4e^{-2} \approx 0.541$

d. $1 - e^{-1} - \frac{7}{2}e^{-2} \approx 0.158$

A Type II error occurs when H_0 is accepted despite the true parameter value being $\lambda = 2$ (i.e. H_1 is true). Under $\lambda = 2$, the independent counts $X_1, X_2 \sim \text{Po}(2)$. The probability mass function is: $P(X = k) = \frac{2^k e^{-2}}{k!}$ Specifically: $P(X = 0) = e^{-2}, \quad P(X = 1) = 2e^{-2}, \quad P(X = 2) = 2e^{-2}$ H_0 can be accepted in two mutually exclusive ways: 1. **In Stage 1:** H_0 is accepted immediately if $X_1 = 0$.

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41 [Challenging] • Target: The discrete uniform distribution • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Calls arrive at a technical support desk according to a Poisson process with an average rate of λ calls per hour. It is known that the probability of receiving exactly 1 call in a given hour is 1.5 times the probability of receiving exactly 3 calls in that same hour. Find the probability that the desk receives exactly 4 calls during a randomly chosen 3-hour period.

a. $54e^{-6} \approx 0.134$ ✓

b. $\frac{2}{3}e^{-2} \approx 0.0902$

c. $\frac{2187}{8}e^{-9} \approx 0.0337$

d. $\frac{625}{24}e^{-5} \approx 0.175$

1. **Find the parameter λ for a 1-hour period:** Let X be the number of calls received in 1 hour. Then $X \sim \text{Po}(\lambda)$. The probability mass function is $\text{P}(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$. We are given: $\text{P}(X = 1) = 1.5\,\text{P}(X = 3)$ Substitute the formula for each probability.

42 [Challenging] • Target: Constructing a probability distribution from a real scenario • Cross-Lesson: 3.01 Graphs & Transformations

The rational function f is defined for all real $xotin\{-1, 2\}$ by $f(x) = \frac{x^3 + kx + 7}{(x+1)(x-2)^2}$ where k is a constant. The partial fraction decomposition of $f(x)$ takes the form $A + \frac{B}{x+1} + \frac{C}{x-2} + \frac{D}{(x-2)^2}$ where $A, B, C,$ and D are real constants. Given that $C = 2$, find the value of k.

a. -3 ✓

b. 1

c. 3

d. 24

Step 1: Determine the polynomial quotient A. The degree of the numerator is 3 and the degree of the denominator is $1 + 2 = 3$. Expanding the denominator gives:
$(x+1)(x-2)^2 = (x+1)(x^2 - 4x + 4) = x^3 - 3x^2 + 4$ Since the leading coefficient of both the numerator and the denominator is 1, polynomial division yields a constant quotient: $A = \frac{1}{1} = 1$ Step 2: Set up the algebraic identity. Multiplying both sides of the partial fraction expansion by $(x+1)(x-2)^2$.

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43 [Challenging] • Target: Cumulative distribution function • Cross-Lesson: 4.01 Coordinate Geometry

The function f is defined for all $x \in \mathbb{R}$ by $f(x) = x^3 - 2x^2 - kx + 12$, where k is a real constant. The Newton-Raphson method is used to find an approximation to a root of the equation $f(x) = 0$. Given that an initial approximation $x_0 = 2$ produces a first approximation $x_1 = 4$, find the value of the constant k and the exact value of the second approximation x_2.

a. $k = 5, \quad x_2 = \frac{28}{9}$ ✓

b. $k = 5, \quad x_2 = \frac{44}{9}$

c. $k = 2, \quad x_2 = \frac{14}{5}$

d. $k = 7, \quad x_2 = \frac{84}{25}$

We solve the problem in two steps: **Step 1: Find the value of k** The Newton-Raphson iteration formula is: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ Differentiating $f(x) = x^3 - 2x^2 - kx + 12$ with respect to x: $f'(x) = 3x^2 - 4x - k$ Evaluating $f(x)$ and $f'(x)$ at $x_0 = 2$: $f(2) = 2^3 - 2(2^2) - 2k + 12 = 8 - 8 - 2k + 12 = 12 - 2k$ $f'(2) = 3(2^2) - 4(2) - k = 12 - 8 - k = 4 - k$ Using the iteration formula for x_1: $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \Rightarrow 4 = 2 - \frac{12 - 2k}{4 - k}$ Rearran.

44 [Challenging] • Target: A well-designed word problem: a raffle ticket game • Cross-Lesson: 5.01 Trigonometry

The function f is defined for $x \in \mathbb{R}$ by $f(x) = 3x + k$, where k is a constant. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x - 4|$. Given that $f^{-1}(4) = 2$, find the value of k and all real solutions to the equation $f^{-1}(x) = g(x)$.

a. $k = -2;\ x = \frac{5}{2},\ x = 7$ ✓

b. $k = -2;\ \backslash;\ x = 7\backslash)$

c. $k = 2;\ x = \frac{7}{2},\ x = 5$

d. $k = -2;\ x = -\frac{5}{2},\ x = 7$

1. **Find the parameter k:** By the definition of an inverse function, $f^{-1}(4) = 2 \Leftrightarrow f(2) = 4$. Substituting $x = 2$ into $f(x)$: $f(2) = 3(2) + k = 6 + k$ $6 + k = 4 \Rightarrow k = -2$ 2. **Determine the inverse function $f^{-1}(x)$:** With $k = -2$, $f(x) = 3x - 2$. Setting $y = 3x - 2$ and solving for x: $3x = y + 2 \Rightarrow x = \frac{y + 2}{3}$ Thus, $f^{-1}(x) = \frac{x + 2}{3}$. 3. **Solve the equation $f^{-1}(x) = g(x)$:** $\frac{x + 2}{3} = |x - 4| \Leftrightarrow x + 2 = 3|x - 4|$.

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45 [Challenging] • Target: Probability distributions • Cross-Lesson: 6.01 Differentiation & Integration

A quality control process monitors the weight, W grams, of chocolate bars produced by a machine. A random sample of 50 chocolate bars was taken, and their weights were recorded. The sample mean weight was 102.5 grams, and the sample standard deviation was 3.5 grams. A) Calculate the standard error of the mean. B) Construct a 95% confidence interval for the mean weight of the chocolate bars. C) The machine is calibrated to produce chocolate bars with a mean weight of $\mu_0 = 100$ grams. Based on your confidence interval, what conclusion can be drawn about the calibration of the machine?

a. The machine is likely calibrated correctly, as μ_0 is within the interval.

b. **The machine is likely not calibrated correctly, as μ_0 is outside the interval.** ✓

c. The machine is likely calibrated correctly, as the sample mean is close to μ_0.

d. The machine is likely not calibrated correctly, as the sample standard deviation is too large.

A) The standard error of the mean (SE) is calculated using the formula: $SE = \frac{s}{\sqrt{n}}$, where s is the sample standard deviation and n is the sample size. Given $s = 3.5$ grams and $n = 50$. $SE = \frac{3.5}{\sqrt{50}} \approx \frac{3.5}{7.071} \approx 0.495$ grams. B) To construct a 95% confidence interval for the mean weight, we use the formula: $CI = \bar{x} \pm z^* \times SE$, where $\bar{x}$ is the sample mean and z^* is the critical z-value for a 95% confidence level. For a 95% confidence lev.

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